Multiply decimals

1 For each of the following products:

i State the number of decimal places in the answer.

ii Calculate the answer.

a 0.3×0.5	b 0.2×0.54	c $0.1 \times 0.2 \times 0.3$	d 0.1×0.05
e 0.03×0.05	f 0.003×0.004	g $0.2 \times 0.5 \times 0.02$	h 0.015×0.2
i 4×0.005	j 54×0.001	k 12×0.02	l 10.3×0.01

2 Evaluate:

a 4×5	b 4×50	c 4×0.5	d 0.4×0.5
e 0.6×0.3	f 2.2×4	g 1.2×3.2	h 4.23×2
i 4.8×8	j 62×9.6	k 3.89×7	l 2.8×6.8

3 For the following products, state whether the answer will be:

- Greater than both numbers.
- Less than both numbers.
- In between the two numbers.

a 3×5	b 3×0.5	c 0.3×0.5	d 7×4
e 0.7×4	f 0.7×0.4	g 15×8	h 15×0.8
i 1.5×0.8			

4 Describe when the product of two numbers is:

a In between the two numbers.

b Less than both of the two numbers.

Area method

5 Find the area of a rectangle with the following dimensions:

a Length of 0.5 cm, width of 0.8 cm

b Length of 0.2 m, width of 0.07 m

6 Consider a rectangle with side lengths of 4.67 and 9 units.

a Complete the table by finding the area of each smaller rectangle as a decimal.

	4	0.6	0.07
9			

b Hence, find 4.67×9 as a decimal.

7 Consider a rectangle with side lengths of 46.2 and 6 units.

a Complete the table by finding the area of each smaller rectangle as a decimal.

	40	6	0.2
6			

b Hence, find 46.2×6 as a decimal.

8 Consider a rectangle with side lengths of 4.92 and 0.6 units.

a Complete the table by finding the area of each smaller rectangle as a decimal.

	4	0.9	0.02
0.6			

b Hence, find 4.92×0.6 as a decimal.

9 Consider a rectangle with side lengths of 2.3 and 3.1 units:

a Complete the table by finding the area of each smaller rectangle as a decimal.

	2	0.3
3		
0.1		

b Hence, find 2.3×3.1 as a decimal.

10 Consider a rectangle with side lengths of 0.87 and 7.8 units.

a Complete the table by finding the area of each smaller rectangle as a decimal.

	0.8	0.07
7		
0.8		

b Hence, find 0.87×7.8 as a decimal.

11 Use the area method to find 0.47×0.89 as a decimal.

12 Use the area method to find 4.86×0.86 as a decimal.

Mixed operations

13 Evaluate:

c $3.2 \times 0.4 + 0.5$

d $0.9 \times 0.6 \times 0.1$

e $1.2 \times 1.1 - 1$

f $3.2 \times 0.5 - 0.7$

g $3.2 + 4.4 \times 0.25$

h $5.6 - 0.8 \times 1.5$

14 Evaluate:

a $(1.1 + 1.2) \times 0.2$

b $(4.2 - 3.6) \times 0.5$

c $(0.1 + 0.2) \times (0.5 - 0.4)$

d $0.5 \times 0.8 - 0.9 \times 0.1$

e $(3.3 + 2.1) \times (0.6 - 0.4)$

f $0.7 \times 0.35 \times 0.2$